

DecodeLabs Internship

Project 1 Report

Project: Advanced Exploratory Data Analysis (EDA) & Feature Engineering

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Objective: Prepare a raw dataset for machine learning by cleaning, transforming and engineering meaningful features.

1. Introduction

Data preprocessing is one of the most important stages in the data science pipeline. Real-world datasets often contain missing values, duplicate records, inconsistent formatting, and statistical outliers. These issues reduce model accuracy and reliability. In this project, a structured preprocessing workflow was implemented to convert the raw dataset into a clean and analysis-ready dataset.

2. Dataset Overview

The provided e-commerce dataset contains customer orders, product details, payment information, discounts, referral sources, quantities, and pricing information. The dataset was inspected using Pandas to understand data types, identify missing values, and examine statistical distributions before preprocessing.

3. Methodology

Step	Description
Data Loading	Imported Excel dataset using Pandas.
Data Inspection	Used info(), describe(), head() and null-value analysis.
Duplicate Removal	Removed duplicate observations.
Missing Values	Filled numeric values with Median and categorical values with Mode.
Outlier Handling	Applied IQR-based clipping to reduce extreme values.
Feature Engineering	Generated additional useful features from existing columns.
Export	Saved final processed dataset as cleaned_dataset.csv.

4. Results & Findings

The preprocessing pipeline successfully removed inconsistencies from the dataset. Missing values were imputed using statistical techniques, duplicate records were eliminated, and outliers were controlled using the Interquartile Range (IQR) approach. The cleaned dataset is more consistent and suitable for visualization, predictive modeling, and business analytics.

5. Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- OpenPyXL
- Jupyter Notebook

6. Conclusion

This project demonstrates the complete preprocessing workflow required before building machine learning models. Proper handling of missing values, duplicate records, outliers, and feature engineering improves data quality and helps generate more accurate and reliable predictive models. The final cleaned dataset is now ready for advanced EDA and machine learning applications.